

Praneesh R V
CB.SC.U4CYS23036

Web Application Security Lab 3

Students.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Activity 1: XML document containing the details of five
students. -->
<students>
  <student>
    <registerNumber>CB22001</registerNumber>
    <name>Alice Akash</name>
    <department>CSE</department>
    <semester>5</semester>
    <cgpa>8.92</cgpa>
  </student>
  <student>
    <registerNumber>CB22002</registerNumber>
    <name>Dinesh Kumar</name>
    <department>CSE</department>
    <semester>5</semester>
    <cgpa>9.14</cgpa>
  </student>
  <student>
    <registerNumber>CB22003</registerNumber>
    <name>Varun S</name>
    <department>IT</department>
    <semester>6</semester>
    <cgpa>8.45</cgpa>
  </student>
  <student>
    <registerNumber>CB22004</registerNumber>
```

```

    <name>Karthik G</name>
    <department>ECE</department>
    <semester>7</semester>
    <cgpa>7.86</cgpa>
  </student>
  <student>
    <registerNumber>CB22005</registerNumber>
    <name>Varunn Nair</name>
    <department>CSE</department>
    <semester>5</semester>
    <cgpa>9.35</cgpa>
  </student>
</students>

```

Students.xsd

```

<?xml version="1.0" encoding="UTF-8"?>
<!--
  Activity 2: schema for students.xml with value restrictions.
    - semester must be an integer between 1 and 8
    - cgpa must be a decimal between 0 and 10
  Extra restrictions kept for data integrity:
    - registerNumber must match the pattern CB + 5 digits
    - department must be one of the known department codes
-->
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
  elementFormDefault="qualified">

  <!-- ===== Simple types (the actual validation rules) =====
-->

  <xs:simpleType name="registerNumberType">
    <xs:restriction base="xs:string">
      <xs:pattern value="CB[0-9]{5}"/>
    </xs:restriction>
  </xs:simpleType>

```

```

        </xs:restriction>
    </xs:simpleType>

    <xs:simpleType name="nameType">
        <xs:restriction base="xs:string">
            <xs:minLength value="1"/>
            <xs:maxLength value="60"/>
        </xs:restriction>
    </xs:simpleType>

    <xs:simpleType name="departmentType">
        <xs:restriction base="xs:string">
            <xs:enumeration value="CSE"/>
            <xs:enumeration value="IT"/>
            <xs:enumeration value="ECE"/>
            <xs:enumeration value="EEE"/>
            <xs:enumeration value="MECH"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- semester: 1 <= value <= 8 -->
    <xs:simpleType name="semesterType">
        <xs:restriction base="xs:integer">
            <xs:minInclusive value="1"/>
            <xs:maxInclusive value="8"/>
        </xs:restriction>
    </xs:simpleType>

    <!-- cgpa: 0 <= value <= 10, at most 2 digits after the
decimal point -->
    <xs:simpleType name="cgpaType">
        <xs:restriction base="xs:decimal">
            <xs:minInclusive value="0"/>
            <xs:maxInclusive value="10"/>

```

```

        <xs:fractionDigits value="2"/>
    </xs:restriction>
</xs:simpleType>

<!-- ===== Structure ===== -->

<xs:element name="students">
    <xs:complexType>
        <xs:sequence>
            <xs:element name="student" maxOccurs="unbounded">
                <xs:complexType>
                    <xs:sequence>
                        <xs:element name="registerNumber"
type="registerNumberType"/>
                        <xs:element name="name"
type="nameType"/>
                        <xs:element name="department"
type="departmentType"/>
                        <xs:element name="semester"
type="semesterType"/>
                        <xs:element name="cgpa"
type="cgpaType"/>
                    </xs:sequence>
                </xs:complexType>
            </xs:element>
        </xs:sequence>
    </xs:complexType>

    <!-- register number must be unique across the document
-->

    <xs:unique name="uniqueRegisterNumber">
        <xs:selector xpath="student"/>
        <xs:field xpath="registerNumber"/>
    </xs:unique>

```

```
</xs:element>

</xs:schema>
```

Validation

XML data to validate

```
28     <department>ECE</department>
29     <semester>7</semester>
30     <cgpa>7.86</cgpa>
31   </student>
32   <student>
33     <registerNumber>CB22005</registerNumber>
34     <name>Varunn Nair</name>
35     <department>CSE</department>
36     <semester>5</semester>
37     <cgpa>9.35</cgpa>
38   </student>
39 </students>
40
```

XML schema (XSD) data

```
70     </xs:element>
71   </xs:sequence>
72 </xs:complexType>
73
74   <!-- register number must be unique across the document -->
75   <xs:unique name="uniqueRegisterNumber">
76     <xs:selector xpath="student"/>
77     <xs:field xpath="registerNumber"/>
78   </xs:unique>
79 </xs:element>
80
81 </xs:schema>
82
```

☒ **Data Security Warning** - Tick this box to confirm you understand all data is stored on our servers for processing and may be retained for service improvement.

Validate

If you have sensitive data, please [Download the Free Liquid Studio Community Edition](#) and use the tools directly from your desktop.

Document Valid

Students.json

```
[
{
  "registerNumber": "CB22001",
  "name": "Alice Akash",
  "department": "CSE",
  "semester": 5,
  "cgpa": 8.92
},
```

```
{
  "registerNumber": "CB22002",
  "name": "Dinesh Kumar",
  "department": "CSE",
  "semester": 5,
  "cgpa": 9.14
},
{
  "registerNumber": "CB22003",
  "name": "Varun S",
  "department": "IT",
  "semester": 6,
  "cgpa": 8.45
},
{
  "registerNumber": "CB22004",
  "name": "Karthik G",
  "department": "ECE",
  "semester": 7,
  "cgpa": 7.86
},
{
  "registerNumber": "CB22005",
  "name": "Varunn Nair",
  "department": "CSE",
  "semester": 5,
  "cgpa": 9.35
}
]
```

Ajax.html

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,
initial-scale=1">
<title>Activity 4 – AJAX (XMLHttpRequest)</title>
<style>
    :root { color-scheme: light; }
    body { font-family: Arial, Helvetica, sans-serif; margin:
2rem;
        background: #fff; color: #111; }
    table { border-collapse: collapse; margin-top: 1rem; }
    th, td { border: 1px solid #666; padding: 6px 12px;
text-align: left; }
    th { background: #eee; }
    .error { color: #b00; font-weight: bold; }
</style>
</head>
<body>

<h2>Activity 4 – Student list loaded with AJAX
(XMLHttpRequest)</h2>

<button onclick="loadStudents()">Load Students</button>

<div id="output"></div>

<script>
function loadStudents() {

    var xhr = new XMLHttpRequest();

    xhr.open("GET", "students.php", true);

    xhr.onload = function () {
```

```

        // A response arrived, but it may still be a 404 or a
500.

        if (xhr.status !== 200) {
            showError("Server responded with status " +
xhr.status + ".");
            return;
        }

        var students = JSON.parse(xhr.responseText);

        var table = "<table>";
        table += "<tr><th>Register
No</th><th>Name</th><th>Department</th>" +
            "<th>Semester</th><th>CGPA</th></tr>";

        for (var i = 0; i < students.length; i++) {
            var s = students[i];
            table += "<tr>";
            table += "<td>" + s.registerNumber + "</td>";
            table += "<td>" + s.name + "</td>";
            table += "<td>" + s.department + "</td>";
            table += "<td>" + s.semester + "</td>";
            table += "<td>" + s.cgpa + "</td>";
            table += "</tr>";
        }

        table += "</table>";

        document.getElementById("output").innerHTML = table;
    };

    // Fires when the request never reaches the server at all.
    xhr.onerror = function () {

```

```

        showError("Unable to reach the server.");
    };

    xhr.send();
}

function showError(message) {
    document.getElementById("output").innerHTML =
        "<p class='error'>" + message + "</p>";
}
</script>

</body>
</html>

```

Activity 4 — Student list loaded with AJAX (XMLHttpRequest)

Load Students

Register No	Name	Department	Semester	CGPA
CB22001	Alice Thomas	CSE	5	8.92
CB22002	Bharath Kumar	CSE	5	9.14
CB22003	Divya Ramesh	IT	6	8.45
CB22004	Karthik Subramanian	ECE	7	7.86
CB22005	Meera Nair	CSE	5	9.35

Fetch.html

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">

```

```
<meta name="viewport" content="width=device-width,
initial-scale=1">
<title>Activity 5 – Fetch API</title>
<style>
    :root { color-scheme: light; }
    body { font-family: Arial, Helvetica, sans-serif; margin:
2rem;
        background: #fff; color: #111; }
    table { border-collapse: collapse; margin-top: 1rem; }
    th, td{ border: 1px solid #666; padding: 6px 12px;
text-align: left; }
    th    { background: #eee; }
    .error{ color: #b00; font-weight: bold; }
</style>
</head>
<body>

<h2>Activity 5 – Student list loaded with the Fetch API</h2>

<button onclick="loadStudents()">Load Students</button>

<div id="output"></div>

<script>
function loadStudents() {

    fetch("students.php")

        .then(response => {
            // fetch() only rejects on a network failure, so HTTP
error
            // codes such as 404 and 500 have to be checked by
hand.
            if (!response.ok) {
```

```

        throw new Error("HTTP " + response.status);
    }
    return response.json();
})

.then(data => {

    let table = "<table>";
    table += "<tr><th>Register
No</th><th>Name</th><th>Department</th>" +
        "<th>Semester</th><th>CGPA</th></tr>";

    data.forEach(student => {
        table += "<tr>";
        table += "<td>" + student.registerNumber +
"</td>";

        table += "<td>" + student.name + "</td>";
        table += "<td>" + student.department + "</td>";
        table += "<td>" + student.semester + "</td>";
        table += "<td>" + student.cgpa + "</td>";
        table += "</tr>";
    });

    table += "</table>";

    document.getElementById("output").innerHTML = table;
})

.catch(error => {
    document.getElementById("output").innerHTML =
        "<p class='error'>Unable to retrieve student
data.</p>";
    console.error(error);
});

```

```
}

</script>

</body>

</html>
```

Activity 5 — Student list loaded with the Fetch API

Load Students

Register No	Name	Department	Semester	CGPA
CB22001	Alice Thomas	CSE	5	8.92
CB22002	Bharath Kumar	CSE	5	9.14
CB22003	Divya Ramesh	IT	6	8.45
CB22004	Karthik Subramanian	ECE	7	7.86
CB22005	Meera Nair	CSE	5	9.35

Student Information Viewer

students.xml → validated against students.xsd → read with simplexml_load_file() → json_encode() → Fetch API → table below. The page is never reloaded.

Load Students

Register No	Name	Department	Semester	CGPA
CB22001	Alice Akash	CSE	5	8.92
CB22002	Dinesh Kumar	CSE	5	9.14
CB22003	Varun S	IT	6	8.45
CB22004	Karthik G	ECE	7	7.86
CB22005	Varunn Nair	CSE	5	9.35

5 student record(s) loaded.

